

Отчёт по лабораторной работе 4

Подготовка экспериментального стенда GNS3

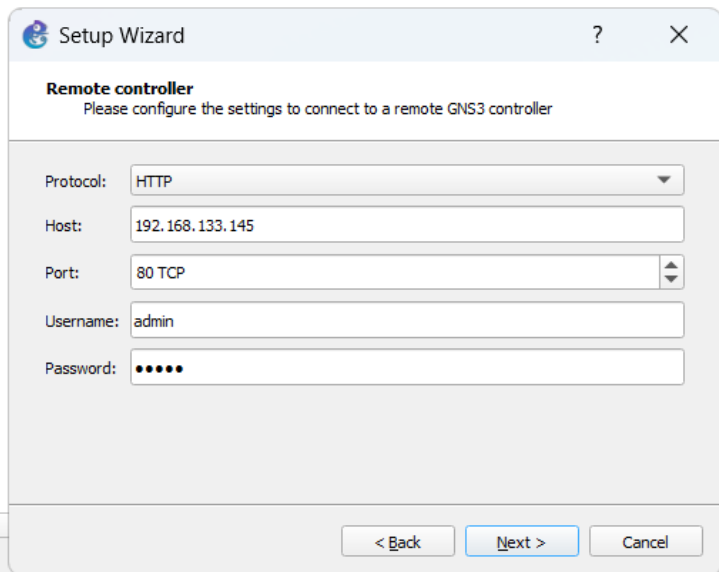
Абрикосов Артём

13 января 2026

Цель работы

Установить и настроить программный комплекс GNS3 и сопутствующее программное обеспечение, а также подготовить лабораторный стенд для моделирования сетевых устройств.

Выполнение работы



The screenshot shows a 'Setup Wizard' window with a title bar containing a question mark and a close button. The main heading is 'Remote controller' with a subtitle 'Please configure the settings to connect to a remote GNS3 controller'. Below this, there are five input fields: 'Protocol' (a dropdown menu set to 'HTTP'), 'Host' (a text box containing '192.168.133.145'), 'Port' (a text box containing '80 TCP' with up and down arrow buttons), 'Username' (a text box containing 'admin'), and 'Password' (a text box with five dots). At the bottom, there are three buttons: '< Back', 'Next >' (highlighted with a blue border), and 'Cancel'.

Setup Wizard ? X

Remote controller
Please configure the settings to connect to a remote GNS3 controller

Protocol: HTTP

Host: 192.168.133.145

Port: 80 TCP

Username: admin

Password: •••••

< Back Next > Cancel

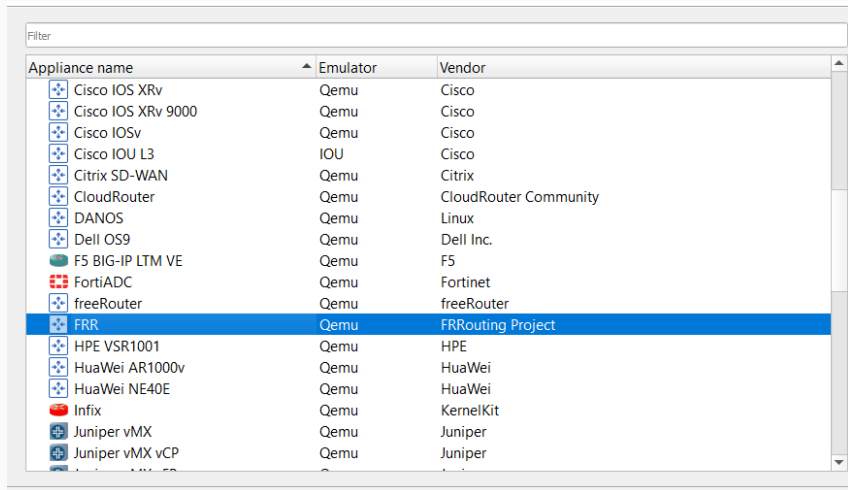
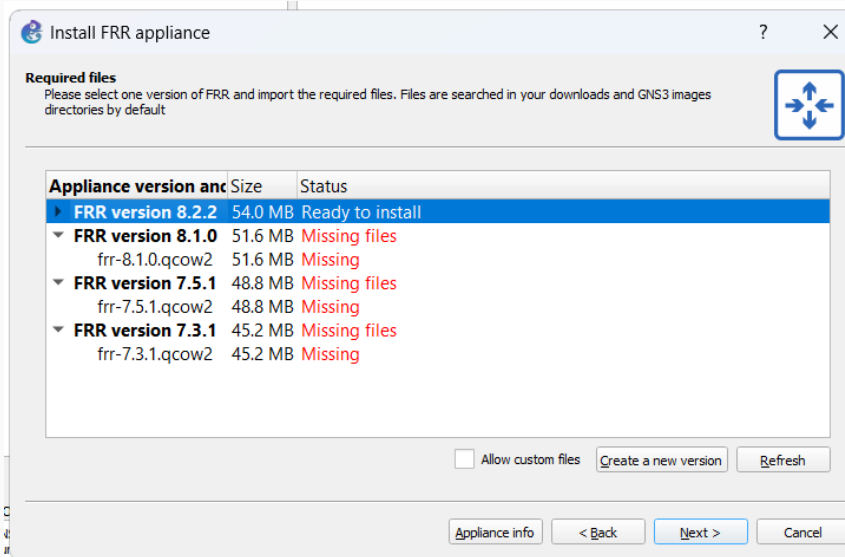
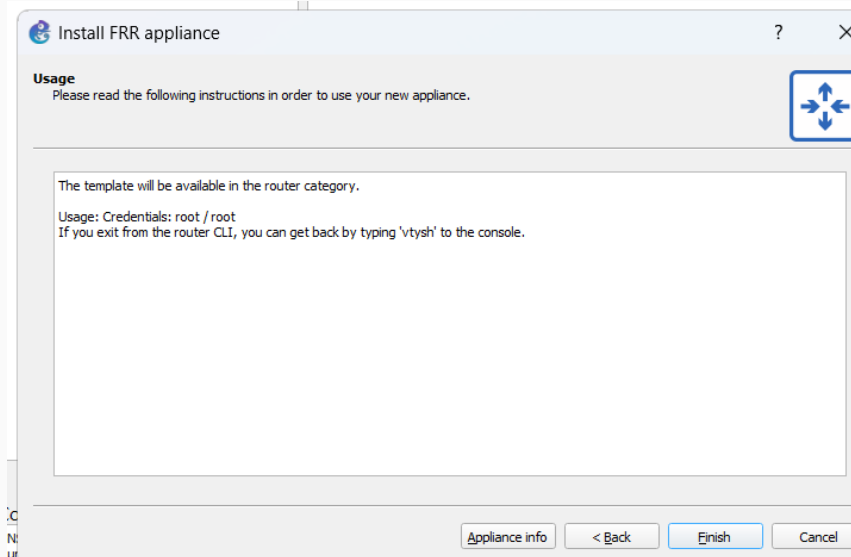
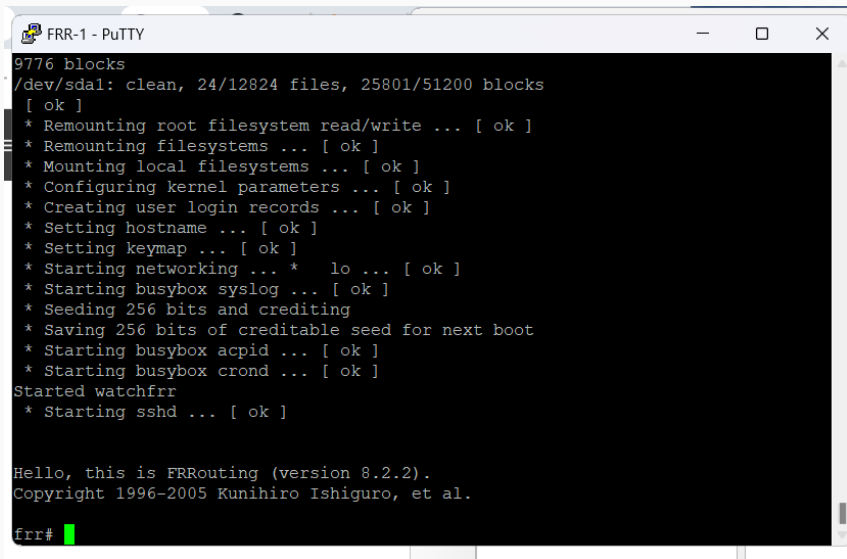


Рис. 2: Выбор образа FRR





Проверка запуска FRR



```
FRR-1 - PuTTY
9776 blocks
/dev/sda1: clean, 24/12824 files, 25801/51200 blocks
[ ok ]
* Remounting root filesystem read/write ... [ ok ]
* Remounting filesystems ... [ ok ]
* Mounting local filesystems ... [ ok ]
* Configuring kernel parameters ... [ ok ]
* Creating user login records ... [ ok ]
* Setting hostname ... [ ok ]
* Setting keymap ... [ ok ]
* Starting networking ... *   lo ... [ ok ]
* Starting busybox syslog ... [ ok ]
* Seeding 256 bits and crediting
* Saving 256 bits of creditable seed for next boot
* Starting busybox acpid ... [ ok ]
* Starting busybox crond ... [ ok ]
Started watchfrr
* Starting sshd ... [ ok ]

Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr#
```

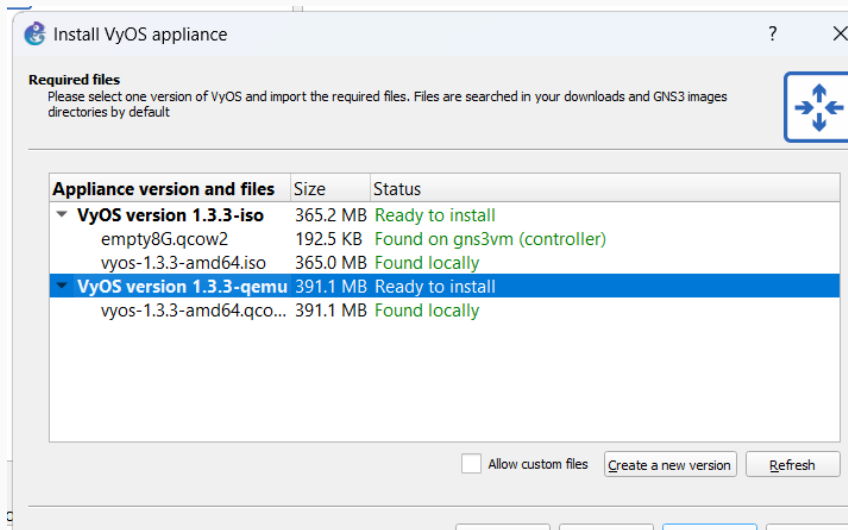
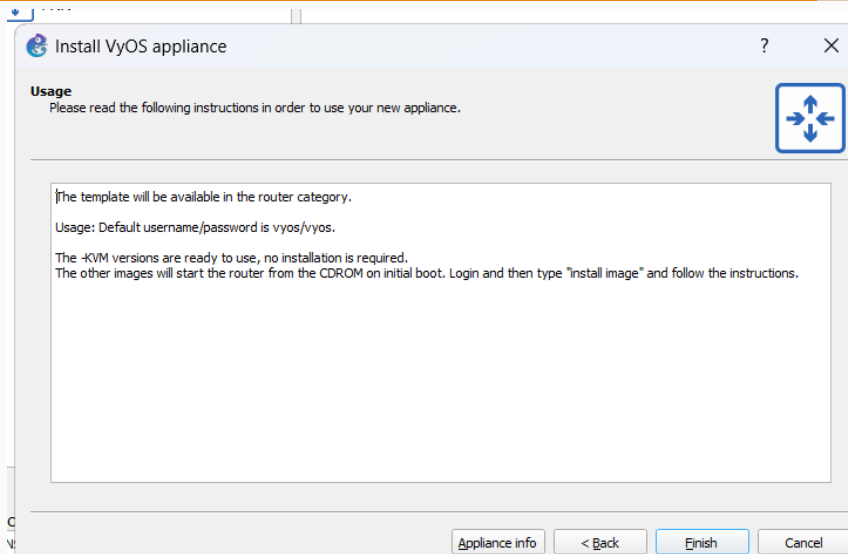
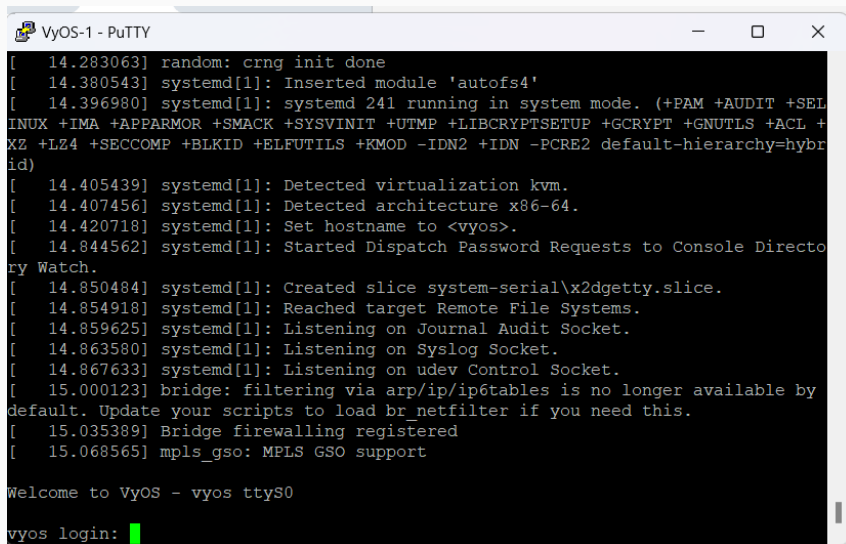


Рис. 6: Выбор версии VyOS



Проверка запуска VyOS



```
[ 14.283063] random: crng init done
[ 14.380543] systemd[1]: Inserted module 'autofs4'
[ 14.396980] systemd[1]: systemd 241 running in system mode. (+PAM +AUDIT +SEL
INUX +IMA +APPARMOR +SMACK +SYSVINIT +UTMP +LIBCRYPTSETUP +GCRYPT +GNUTLS +ACL +
XZ +LZ4 +SECCOMP +BLKID +ELFUTILS +KMOD -IDN2 +IDN -PCRE2 default-hierarchy=hybr
id)
[ 14.405439] systemd[1]: Detected virtualization kvm.
[ 14.407456] systemd[1]: Detected architecture x86-64.
[ 14.420718] systemd[1]: Set hostname to <vyos>.
[ 14.844562] systemd[1]: Started Dispatch Password Requests to Console Directo
ry Watch.
[ 14.850484] systemd[1]: Created slice system-serial\x2dgetty.slice.
[ 14.854918] systemd[1]: Reached target Remote File Systems.
[ 14.859625] systemd[1]: Listening on Journal Audit Socket.
[ 14.863580] systemd[1]: Listening on Syslog Socket.
[ 14.867633] systemd[1]: Listening on udev Control Socket.
[ 15.000123] bridge: filtering via arp/ip/ip6tables is no longer available by
default. Update your scripts to load br_netfilter if you need this.
[ 15.035389] Bridge firewalling registered
[ 15.068565] mpls_gso: MPLS GSO support

Welcome to VyOS - vyos ttyS0

vyos login: █
```

Рис. 8: Загрузка маршрутизатора VyOS

Итоги работы

В ходе лабораторной работы выполнена установка и настройка GNS3 и GNS3 VM, а также произведён импорт и проверка работоспособности маршрутизаторов FRR и VyOS. Подготовленный лабораторный стенд готов к дальнейшему моделированию и исследованию сетевых конфигураций.